Gunhee Lee

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Research Interests

World Model, Generative AI, Content Intelligence, Computer Vision, Machine Learning

Education

Pohang University of Science and Technology (POSTECH) *South Korea*
M.S. in Computer Graphics (Aug. 2017 - Aug. 2019)
Thesis: Deep Image Recolorization using Histogram Analogy

Handong Global University (HGU) *South Korea*
B.S. in Computer Science; Summa cum laude (GPA 4.08/4.50) (Mar. 2011 - Aug. 2017)

Professional Experience

NC AI *South Korea*
Generative AI Research Engineer at Visual Contents Service Team (Feb. 2025 - Present)
NCSOFT *South Korea*
Generative AI Research Engineer at Image Generation Team (Jan. 2020 - Feb. 2025)

- **AI SaaS Platform (VARCO) for Diffusion-based Image Generation** [Link1][Link2]
  - Conducted research on developing base foundational models and corresponding adaptors.
  - Spearheaded the development of virtual try-on technology using LoRA-based generative model fine-tuning with multi-GPU training.
  - Designed and implemented multi-view image generation features, integrating data refinement, model training, and AI architecture pipeline design.
  - Led the development of the volume generation feature, enabling lighting adaptation and automatic coloring.
- **Controllable Image Generation Research**
  - Designed a coarse-to-fine conditional pipeline and model architecture for high-resolution landscape image synthesis, enabling fast, user-controlled, and highly detailed visuals using GANs.
  - Built an automated synthetic face dataset pipeline that generates datasets from single face images through latent manipulation in GANs.
- **Advanced Video FaceSwap Research**
  - Developed high resolution (1024px) video face-swap to preserve identity for digital human project.
  - Developed video head-swap application utilizing single face image input, enabling user control.
- **Realistic Video Talking-Heads Research**
  - Optimized landmark-based GAN models to ensure stable lip-sync animations for an entertainment platform.
  - Constructed specialized datasets for Korean-speech dataset, achieving performance improvements through single- and multicharacter model training.

Selected Publications

- **Gunhee Lee**, Jonghwa Yim, Chanran Kim, Minjae Kim. *StyLandGAN: A StyleGAN-based Landscape Image Synthesis using Depth-map.* AI for Content Creation Workshop (CVPRW), 2022.
- Junyong Lee, Hyeongseok Son, **Gunhee Lee**, Jonghyeop Lee, Sunghyun Cho, Seungyong Lee. *Deep color transfer using histogram analogy.* The Visual Computer (CGI), 2020.
- Hyeongseok Son, **Gunhee Lee**, Sunghyun Cho, Seungyong Lee. *Naturalness-Preserving Image Tone Enhancement Using GANs.* Computer Graphics Forum (Pacific Graphics), 2019.